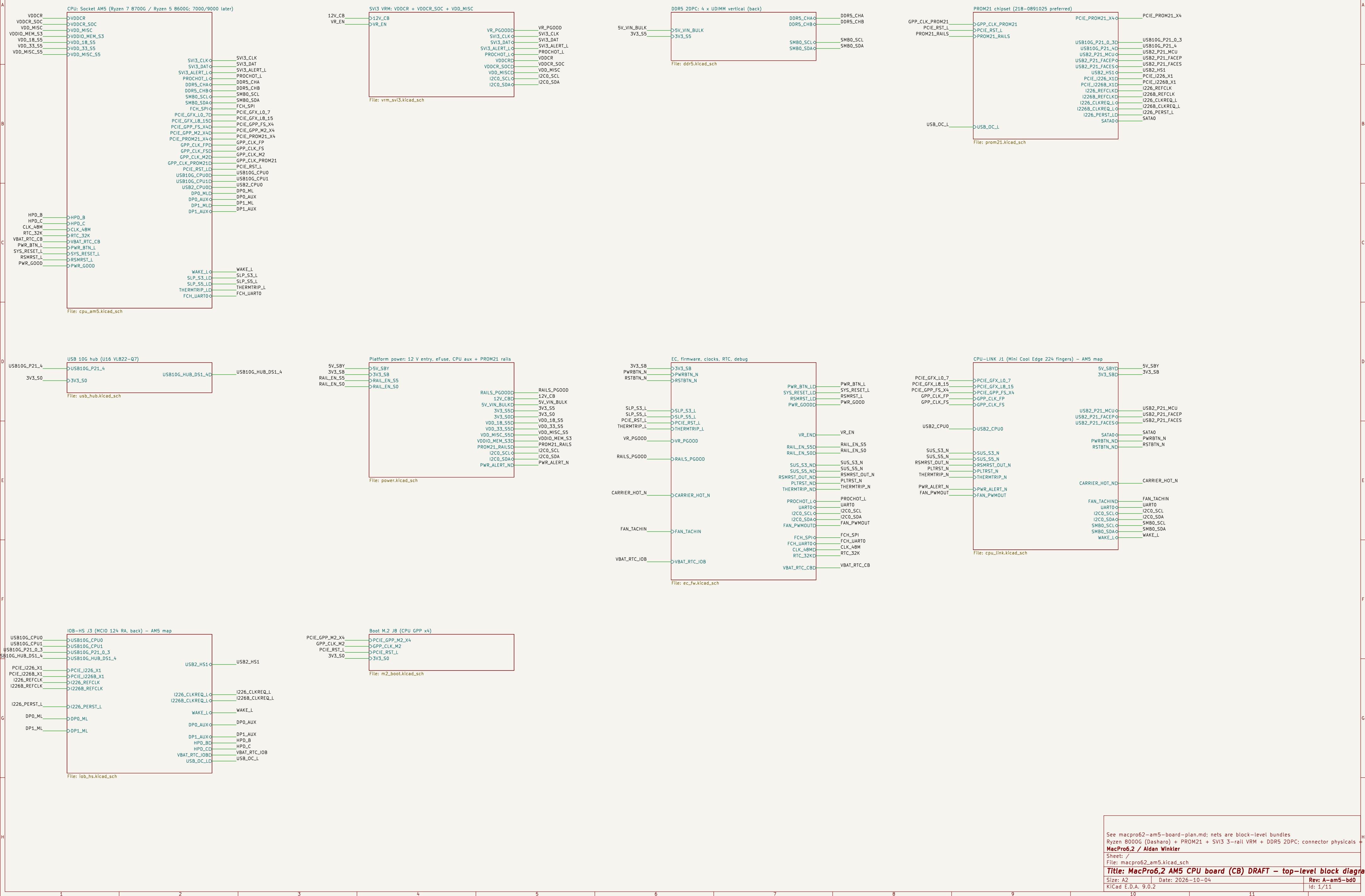


MP62 AM5 CPU BOARD rev A–am5 DRAFT – BLOCK DIAGRAM. Sheets are stubs (labels only). Same label name = same net (bundle).

Off–board: J1 CPU–LINK (card edge –> BP) | J3 IOB–HS MCIO 124 | J8 M.2 boot | LU1/LU2 12 V | J5 debug | J4 TPM. AMD pin names (NDA) replaced by functional names.



BLOCK STUB (rev A block diagram). Contents planned:

- U1 Socket AM5 LGA1718 (Foxconn PE17181/PE17186 or Lotes AZIFS055), MP62 contact frame on the 69.5 x 55 core holes (no stock ILM)
- Primary CPU: Ryzen 8000G (Phoenix) – the only AM5 family Dasharo (coreboot + openSIL) boots today; cTDP 45 W (PPT ~61 W)
- 8000G PCIe 4.0: 20 lanes / 16 usable = GFX x8 (Face P) + GPP x4 (Face S) + GPP x4 (M.2 boot) + x4 PROM21 uplink
- GFX lanes 8–15 are routed to J1 but unused by 8000G; live with Ryzen 7000/9000 (x16 Gen5, AC caps fitted)
- USB: 2 x USB 10G → J3 C1/C2, 1 x USB2 → CPU–LINK SPARE; 2 x USB4 unused in rev A (option: run them at 10G instead of the hub [firmware risk])
- Display: DP0 4–lane → J3 DDIB, DP1 2–lane → J3 DDIC (AUX AC–coupled, HPD from the IOB); 7000/9000 iGPU = 2 lanes class TBD
- FCH (in the CPU): GPP_CLK reflcks (closes U–12 on AM5), SPI ROM, SMBus0 (SPD), SLP_S3_L/SLP_S5_L, RSMRST_L, PWR_GOOD, PCIE_RST_L, THERMTRIP_L, PROCHOT_L
- Pin names are functional placeholders: the AM5 pin list/land map is AMD NDA

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC see	
Sheet: /CPU: Socket AM5 (Ryzen 7 8700G / Ryzen 5 8600G; 7000/9000 later)/ File: cpu_am5.kicad_sch	
Title: MP62 AM5 CPU board: CPU: Socket AM5 (Ryzen 7 8700G / Ryze	
Size: A4	Date: 2026–10–04
KiCad E.D.A. 9.0.2	Rev: A–am5–bd0 Id: 2/11





Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /DDR5 2DPC: 4 x UDIMM vertical (back)/
File: ddr5.kicad_sch

Title: MP62 AM5 CPU board: DDR5 2DPC: 4 x UDIMM vertical (back)

Size: A4Date: 2026-10-04Rev: A-am5-bd0

KiCad E.D.A. 9.0.2Id: 4/11



123456

A

B

C

D

123456

USB10G_P21_4

USB10G_HUB_DS1_4

3V3_S0

123

USB10G_P21_4

USB10G_HUB_DS1_4

3V3_S0

BLOCK STUB (rev A block diagram). Contents planned:

– U16 VIA Labs VL822–Q7 USB 3.2 Gen2 1:4 hub, QFN–76 9x9 (LCSC C42419379); alt Genesys GL3590

– Upstream = PROM21 10G #4; downstream 1–4 = J3 USB3_A1..A4 (the IOB redriver side unchanged); A1–A4 share 10 Gb/s

– Needed because 8000G (2 x 10G) + PROM21 (6 x 10G) = 8 native 10G ports, the IOB needs 10 (C1–C6 + A1–A4)

– Own SPI flash + 25 MHz crystal; bypass option with Ryzen 7000/9000 (4 native 10G) TBD

Block stub: hierarchical labels + one non–BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /USB 10G hub (U16 VL822–Q7)/
File: usb_hub.kicad_sch

Title: MP62 AM5 CPU board: USB 10G hub (U16 VL822–Q7)

Size: A4

Date: 2026–10–04

Rev: A–am5–bd0

KiCad E.D.A. 9.0.2

Id: 6/11

123456

The screenshot displays the KiCad schematic editor interface. On the left, a hierarchical block diagram stub is shown, listing various pins and their connections. The pins are organized into groups, with some pins having multiple connections indicated by numbers in red boxes. The right pane contains a title block and project information.

Pins and Connections:

Pin Name	Connection
3V3_SBD	3V3_SB
PWRBTN_ND	PWRBTN_N
RSTBTN_ND	RSTBTN_N
PWR_BTN_L	PWR_BTN_L
SYS_RESET_L	SYS_RESET_L
RSMRST_L	RSMRST_L
PWR_GOOD	PWR_GOOD
SLP_S3_LD	SLP_S3_L
SLP_S5_LD	SLP_S5_L
PCIE_RST_LD	PCIE_RST_L
THERMTRIP_LD	THERMTRIP_L
VR_EN	VR_EN
VR_PGOOD	VR_PGOOD
RAIL_EN_S5	RAIL_EN_S5
RAIL_EN_S0	RAIL_EN_S0
RAILS_PGOOD	RAILS_PGOOD
SUS_S3_N	SUS_S3_N
SUS_S5_N	SUS_S5_N
RSMRST_OUT_N	RSMRST_OUT_N
PLTRST_N	PLTRST_N
THERMTRIP_N	THERMTRIP_N
CARRIER_HOT_ND	CARRIER_HOT_N
PROCHOT_L	PROCHOT_L
UART0	UART0
I2C0_SCL	I2C0_SCL
I2C0_SDA	I2C0_SDA
FAN_PWMOUT	FAN_PWMOUT
FAN_TACHIN	FAN_TACHIN
FCH_SPI	FCH_SPI
FCH_UART0	FCH_UART0
CLK_48M	CLK_48M
RTC_32K	RTC_32K
VBAT_RTC_IOB	VBAT_RTC_IOB
VBAT_RTC_CB	VBAT_RTC_CB

BLOCK STUB (rev A block diagram). Contents planned:

- U9 RP2350 EC (as LGA1700): AM5 sequencing RSMRST_L -> PWR_BTN_L -> SLP_S5/S3 -> rails -> PWR_GOOD; VR_EN / VR_PGOOD
- Buffers to CPU-LINK (names unchanged): SUS_S3# = SLP_S3_L, SUS_S4-S5# = SLP_S5_L, PLTRST# = PCIE_RST_L, THERMTRIP#, RSMRST_OUT#
- LPT actuator on AM5: PPT/TDC/EDC via the SMU (Dasharo/openSIL interface TBD) - fallback fixed cTDP 45 W + PROCHOT_L (CARRIER_HOT#)
- U10 FCH SPI ROM 32 MB (Dasharo coreboot + openSIL + PSP blobs), external programming; J4 SPI TPM header (fTPM default)
- Y1 48 MHz [Unverified] + 32.768 kHz; VBAT_RTC: IOB coin cell (J3 B26) diode-OR 3V3_S5 -> CPU RTC well; BT1 DNP; J5 debug (FCH UART, SWD, POST)

Title Block:

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC se		
Sheet: /EC, firmware, clocks, RTC, debug/ File: ec_fw.kicad_sch		
Title: MP62 AM5 CPU board: EC, firmware, clocks, RTC, debug		
Size: A4	Date: 2026-10-04	Rev: A-am5-bd0
KiCad E.D.A. 9.0.2		Id: 8/11

